

DECOMPOSER BIOLOGY · CROP RESIDUE AFTER HARVEST

Mycro-Tec® Digester



Selected decomposer organisms for post-harvest residue.

Four selected decomposer organisms, applied once onto crop residue after harvest. They begin the breakdown of stubble, prunings and crop trash so the carbon and nutrients held in it return to the soil ahead of the next crop.

01
Used on
Crop residue in vegetables, orchards and broadacre.

02
Applied as
One spray onto the residue after harvest, with an ammonium sulphate starter.

03
Contains
Three fungi and one actinomycete, declared on the label.

USING DIGESTER

One application after harvest, as close to rain as you can.

Sprayed onto the residue with an ammonium sulphate starter, after harvest and close to rain, before or after it. Boomspray or orchard sprayer. Label rates are on the back page.



01

Vegetables

Crop trash left after harvest



02

Orchards

Prunings and leaf litter on the orchard floor. Not for nut trees.



03

Broadacre

Cereal, rice, canola and legume stubble

WHERE IT PAYS OFF

When it is worth doing.

If you run a nutrition program with us, Digester is the post-harvest step and the prescription blend at establishment follows it. If you don't, it stands on its own after harvest.

01

Residue that will still be there at sowing

Cereal, canola, legume, vegetable or orchard residue that would otherwise carry intact into the next crop.

02

Heavy or lignified loads

Big stubble loads, or material that has sat through a dry summer and stalled.

03

A moisture window you can hit

Applied soon after harvest and as close to rain as you can, before or after it. The breakdown runs on moisture in the residue.

04

Where the basics are already in hand

Residue breaks down fastest in a block where water, soil balance and nutrition are being managed. Digester is a step in that program, not a substitute for it.

Hot, dry spells

Activity falls away through prolonged heat and dryness. Keep out of the heat of the day and line the application up with rain.

Time in the tank

Mix and apply the same day. Overnight in nutrient the fungi grow mycelia and block nozzles.

Nitrogen on its own

Nitrogen feeds the biology already there; it does not add decomposers. The 8 kg of ammonium sulphate is the starter. Do not add more nitrogen to the mix.

WHAT IS IN IT

Four organisms, declared on the label.

Between them they work on every structural part of crop residue. Colony-forming units per millilitre at production, as declared on the label. Three fungi and one actinomycete.

01

Trichoderma harzianum

FUNGUS

$>2\times10^{8}$

CFU/ML

Fast fungal coloniser of cellulose-rich residue; its enzymes begin opening the fibre.

02

Aspergillus niger

FUNGUS

$>2\times10^{7}$

CFU/ML

Produces organic acids and enzymes that work on the tougher, more lignified material.

03

Streptomyces lydicus

ACTINOMYCETE

$>2\times10^{7}$

CFU/ML

Bacterial decomposition alongside the fungi, so the breakdown does not rely on one pathway.

04

Penicillium radicum

FUNGUS

$>1\times10^{7}$

CFU/ML

The third fungus in the mix. Works across the residue surface and the damp base alongside Trichoderma.

WHAT THEY ACT ON

Cellulose

The main plant fibre

Hemicellulose

The supporting fibre network

Lignin

The hard, resistant structure

Pectin

The material that binds plant cells

HOW IT WORKS

How residue breaks down.

The organisms do the work. Moisture and time set the pace.



01

Establishes at the base

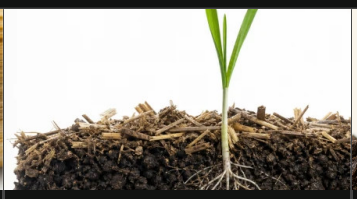
The organisms take hold first where residue meets soil and moisture is held, then spread through damp stubble.



02

Opens the structure

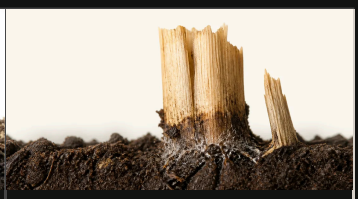
Their enzymes and organic acids work on cellulose, hemicellulose, lignin and pectin, so the carbon locked in the fibre becomes available.



03

Breakdown continues

The selected organisms and the soil's own community keep working through the material ahead of the next crop.



04

How you check it

Digestion starts at the stubble base. Treated residue pulls away from the soil more easily and the stalks soften and weaken as it runs. One field check, no lab needed.

The roles described here are the known behaviour of these organisms on crop residue. They are not a measured result for Digester in a particular paddock or season.

REFERENCE

Rates, mixing and handling.

Label rates and the handling that keeps the biology alive between the drum and the paddock.

CROP GROUP	DIGESTER	STARTER	WATER
01 **Vegetables** Crop trash left after harvest	**1–2 L/ha** up to 3 L/ha heavy residue	**8 kg/ha** every application	**100 L/ha** 1:50 product : water
02 **Orchards** Prunings and leaf litter on the orchard floor. Not for nut trees.	**1–2 L/ha** up to 3 L/ha heavy residue	**8 kg/ha** every application	**100 L/ha** 1:50 product : water
03 **Broadacre** Cereal, rice, canola and legume stubble	**1–2 L/ha** up to 3 L/ha heavy residue	**8 kg/ha** every application	**100 L/ha** 1:50 product : water

Label rates. The ammonium sulphate stays at 8 kg/ha whatever the residue load. One application after harvest; not for in-crop use.

01

Mixing

Shake the drum well. Add Digester to cool, clean, non-chlorinated water first (neutral pH, not below 5.5), then add the ammonium sulphate and stir thoroughly.

Strain as you pour. Keep nozzles and filters at 150 µm or larger.

Mix and apply the same day. Left in the tank overnight it grows mycelia and blocks nozzles.

Digester, water and ammonium sulphate only. Do not swap the starter for urea, fish or anything else.

02

Storage

Store at 5–40 °C in a clean, dry place out of direct sunlight, drums upright and closed. Do not freeze.

Twelve months from the date of manufacture. Once a drum is open, use it rather than keep it: the organisms start to germinate and expire without nutrient. Never store the diluted mix.

Growth on top of the liquid is mycelia, a sign the biology is alive. Shake and use as normal.

03

Safety

Not classified as hazardous. Keep out of reach of children; may irritate skin or eyes, so wash after handling.

Wear gloves and eye protection when mixing. Follow the current label and safety data sheet.

Not certified organic. Allowable-input status is assessed case by case with your certifier.

TALK TO US

Talk to us about Digester.

Tell us the crop, the residue load and when you expect rain. Your Hybrid-Ag agronomist can advise on timing and rate. Label and safety data sheet available on request.



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Mycro-Tec® Digester is made in Australia by Great Southern Biology and supplied by Hybrid-Ag.